

Lab Assignment 9: Quantum Crptography and Quantum Search

Experiment 1: Quantum Key Distribution

Implement the QKD scheme using Qiskit. Refer to <https://github.com/Qiskit/textbook/blob/main/notebooks/ch-algorithms/quantum-key-distribution.ipynb>

Experiment 2: BB84

Implement the BB84 scheme using Qiskit. Refer to <https://github.com/qmunitytech/Tutorials/blob/main/intermediate/The%20BB84%20Quantum%20Cryptography%20algorithm.ipynb>

Experiment 3: Grover's Search

Implement the Grover's search algorithm using Qiskit. Refer to <https://github.com/Qiskit/textbook/blob/main/notebooks/ch-algorithms/grover.ipynb>